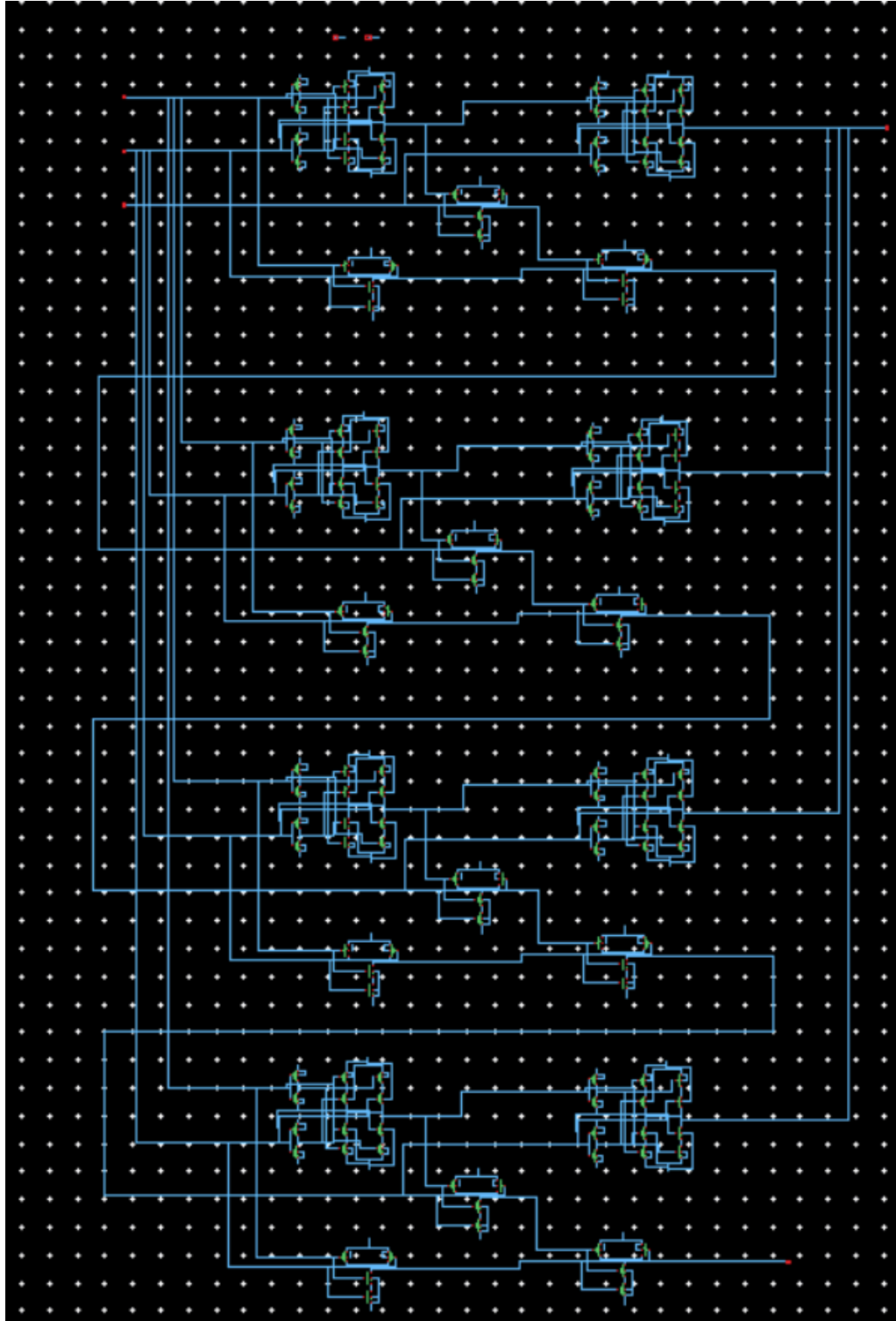


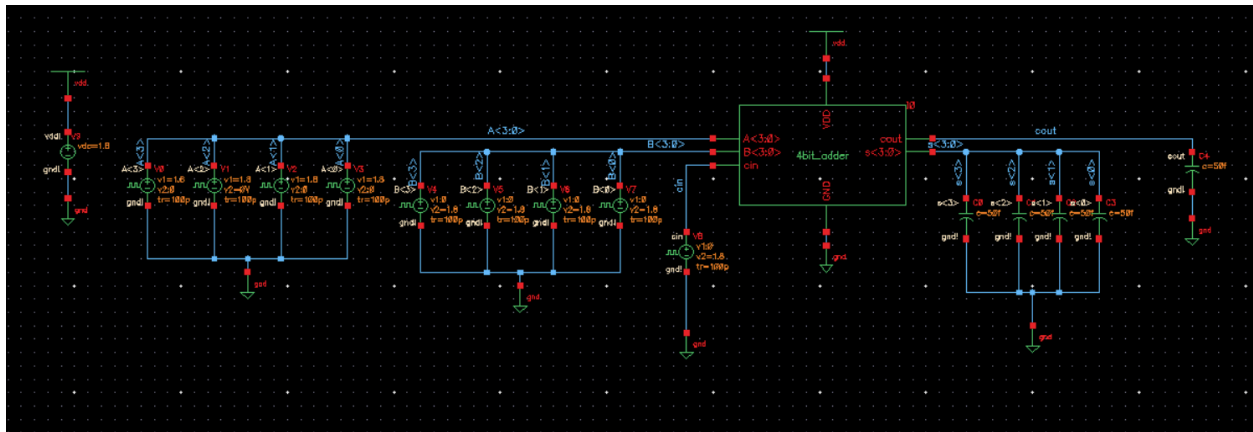
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Lab 9: Optimization using Logical Effort

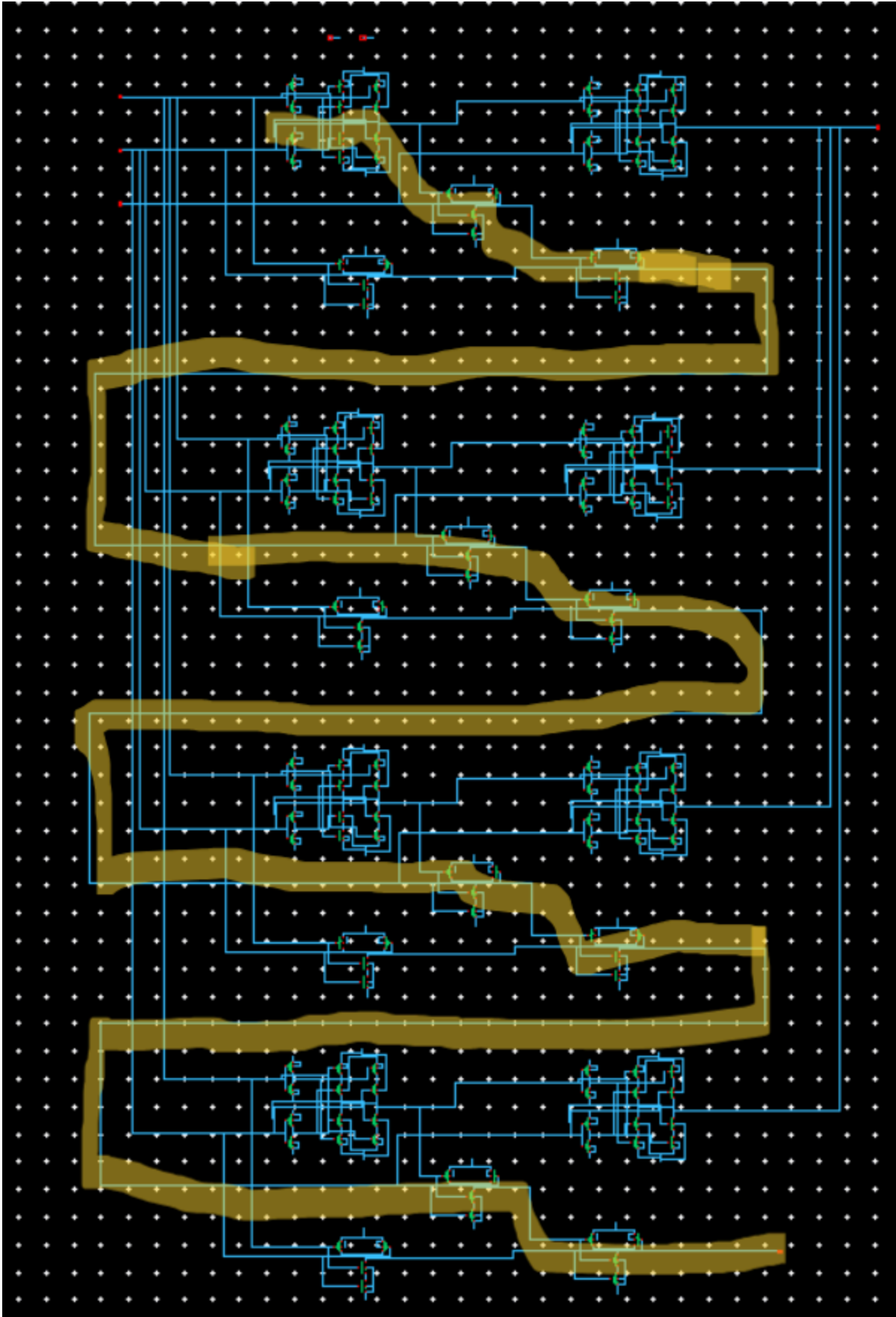
Schematic of 4-bit adder



Simulation Schematic of 4-bit adder



1. Circuit with critical path (2)



2. Calculate G, B, H and attach the calculation procedure (2*3)

Lab 9

$C_{xor} = 6.38e-09$ $C_{nand2} = 3.28e-09$

$G = \pi g_i$ for critical path

$g_{nand} = 4/3$ $g_{xor} = 4$

logical effort

$G = g_{xor} \cdot g_{nand}^8 = 4 \left(\frac{4}{3}\right)^8 = 39.95$

• Compute $H = \text{Path critical effort} = \text{Cout-capacitance} / \text{Cin-capacitance} = C_{load} / C_{xor}$
 $C_{load} = 30 \text{ fF}$, C_{xor} is value from Lab 3

$H = \frac{\text{Cout-capacitance}}{\text{Cin-capacitance}} = \frac{C_{load}}{C_{xor}} = \frac{30 \text{ fF}}{6.38 \times 10^{-9}} = 4.702 \times 10^{-6}$

• $B = \text{Path Branching Effort} = \pi b_i$ ($i = 1, 2, 3, 4, \dots, 9$)

$b_i = (\text{Con-critical-path} + \text{Coff-critical-path}) / \text{Con-critical-path}$

$b_i = \frac{C_{xor} + C_{nand2}}{C_{xor}} \cdot \frac{C_{xor} + C_{nand2}}{C_{nand2}} = 4 \quad i = 1, 2, 3$

$b_i = \left(\frac{C_{xor} + C_{nand2}}{C_{nand2}} \right)^3 = 8.01 \quad i = 4, 5, 6, 7, 8, 9$

$B = \pi b_i = 1.69 \times 10^7$

$f = (GBH) = 2.4495$

$\rightarrow (39.95) \cdot (4.702 \times 10^{-6}) \cdot (1.69 \times 10^7)$

3. Resizing procedure & transistors' sizes (2)

Since, f is less than 2.7, new $f = 3.0$

Resizing and transistor's sizes

Stage 9, last stage with NAND Gate

$$\text{PMOS new size} = \frac{0.85}{0.4} \times \frac{\text{new } C_{in}}{\text{old } C_{in}}$$

$$\text{new } C_{in} = \frac{g_i (C_{out(i)})}{3.0}$$

$$\text{old } C_{in} = \frac{g_i (C_{out(i)})}{2.44}$$

$$\text{PMOS new size} = \frac{0.85}{0.4} \times \frac{\text{new } C_{in}}{\text{old } C_{in}} = 0.85 \mu\text{m}$$

$$\text{NMOS new size} \rightarrow \frac{0.4}{0.85} \times \text{PMOS new size} = 0.4 \mu\text{m}$$

$$\text{Stage 8: PMOS} = 0.9 \mu\text{m} \quad \text{NMOS} = 0.4 \mu\text{m}$$

$$\text{Stage 7: PMOS} = 0.859 \mu\text{m} \quad \text{NMOS} = 0.4 \mu\text{m}$$

$$\text{Stage 6: PMOS} = 0.896 \mu\text{m} \quad \text{NMOS} = 0.42 \mu\text{m}$$

$$\text{Stage 5: PMOS} = 1.015 \mu\text{m} \quad \text{NMOS} = 0.45 \mu\text{m}$$

$$\text{Stage 4: PMOS} = 1.084 \mu\text{m} \quad \text{NMOS} = 0.510 \mu\text{m}$$

$$\text{Stage 3: PMOS} = 1.141 \mu\text{m} \quad \text{NMOS} = 0.537 \mu\text{m}$$

$$\text{Stage 2: PMOS} = 1.352 \mu\text{m} \quad \text{NMOS} = 0.636 \mu\text{m}$$

$$\text{Stage 1: PMOS} = 1.269 \mu\text{m} \quad \text{NMOS} = 0.564 \mu\text{m}$$